



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (14/09/2026 – 20/09/2026)

WEEK 8

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

**DESIGN AND IMPLEMENT A CHATBOT THAT CAN ANSWER TO
DIFFERENT TYPE OF QUERY LIKE STRUCTURED, UNSTRUCTURED AND
SEMANTIC KNOWLEDGE BY DECIDING OWN WHAT TYPE OF QUERY BY**

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COLLEGE NAME :

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14/09/2026	- Designed the overall GenAI conversational data platform architecture with separate agent, ingestion, storage, tools, and evaluation components. - Set up the initial project structure and implemented the LangGraph-based agent workflow, including routing, planning, tool execution, auditing, and answer synthesis.
15/09/2026	- Developed the document ingestion pipeline with document parsing, chunking, metadata enrichment, and knowledge-graph extraction. - Implemented the storage layer with tenant-specific vector, SQL, and graph storage components.
16/09/2026	- Integrated specialist tools for document RAG, SQL, graph queries, trend analysis, vision, and web search. - Added multi-tenant demo data for two different organizations to test domain-agnostic data handling.
17/09/2026	- Implemented evaluation components for routing accuracy, retrieval performance, LLM-based judging, latency tracking, and red-team testing. - Added small, medium, and large test workflows along with demo query scripts for validating the platform.
18/09/2026	- Configured the platform to use Ollama as a free local LLM and adapted the LLM configuration for local inference. - Set up the project for Windows/VS Code and resolved initial environment and configuration issues during local testing.
19/09/2026	- Ran the initial end-to-end test of the document ingestion and RAG pipeline and fixed the local LLM model configuration issue. - Successfully validated document ingestion, knowledge-graph extraction, query routing, retrieval, and answer generation using the local LLM.
20/09/2026	Holiday (Sunday)

Next Week Plan	• Improve query routing using the expanded evaluation dataset.
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REFERENCE:

https://microsoft.github.io/graphrag/get_started/

https://microsoft.github.io/graphrag/index/default_dataflow/

<https://docs.ollama.com/>

<https://www.sqlite.org/docs.html>

<https://github.com/meta-llama/llama3>

<https://docs.python.org/3/library/sqlite3.html>

<https://www.postgresql.org/docs/>

<https://fastapi.tiangolo.com/tutorial/>

<https://docs.langchain.com/oss/python/langgraph/overview>

SCREEN SHOTS:

